

# SELECTING A PUMP FOR CONTINUOUS DUTY

## PUMP APPLICATION GUIDE

A decision framework for turning a real process duty into a pump package that can be specified, commissioned, and maintained. Start with the operating profile, not a catalogue point.

| PRIMARY AUDIENCE                                         | REFERENCE STATUS             |
|----------------------------------------------------------|------------------------------|
| Process engineers / Package engineers / Operations teams | Illustrative project concept |

## FIELD SEQUENCE

| ENGINEERING LENS                 | WHAT TO RESOLVE                                                                                                                                                                                     |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01 / Define the duty             | Record normal, minimum, maximum, and upset flow; differential pressure; fluid temperature; density and viscosity; operating hours; start and stop frequency; and the consequence of losing flow.    |
| 02 / Build the system curve      | Account for static head, pipework, fittings, control valves, filters, heat exchangers, elevation, and the expected operating range. Keep design assumptions visible so they can be challenged.      |
| 03 / Check suction conditions    | Review available NPSH across temperature and flow cases. Include upstream losses, tank level, vapor pressure, strainers, and transient conditions rather than relying on one steady-state estimate. |
| 04 / Match materials and sealing | Confirm compatibility with concentration, solids, temperature, corrosivity, and cleaning regime. Select the seal and bearing approach with the maintenance plan in view.                            |
| 05 / Close the package           | Coordinate driver, controls, minimum-flow protection, instruments, isolation, baseplate, access, test basis, spares, and commissioning data as one package.                                         |

# SELECTING A PUMP FOR CONTINUOUS DUTY / HANDOVER RECORD

## PUMP APPLICATION GUIDE

A useful field guide ends with evidence that the system is ready for the next team. Capture the records below in the project handover pack and keep exceptions visible until they are closed.

## RECOMMENDED RECORDS

|    |                                      |
|----|--------------------------------------|
| 01 | Duty and upset case table            |
| 02 | System curve assumptions             |
| 03 | NPSH and suction review              |
| 04 | Materials and sealing basis          |
| 05 | Package and commissioning data sheet |

### REFERENCE NOTE

This guide supports early planning and field conversations. It does not replace the approved drawings, equipment instructions, site procedures, inspection plan, or project acceptance criteria.